COURSE CODE:ITA0608

COURSE NAME: Machine learning

EXPERIMENT-5: Experiment: Implementation of K-Nearest Neighbours (K-NN) in Python

Aim: To implement the K-Nearest Neighbours (K-NN) classification algorithm in Python and classify data based on the nearest neighboring samples.

Algotithm:

1. Import the required Python libraries.
2. Load the dataset using Pandas.
3. Separate input features (X) and output labels (y).
4. Split the dataset into training and testing sets.
5. Create a K-Nearest Neighbours classifier with K = 3.
6. Train the classifier using the training data.
7. Predict the class labels for the test data.
8. Calculate the accuracy of the classifier.
9. Display the predicted values and accuracy.

Code:

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

data = pd.read_csv("knn_dataset.csv")

X = data.iloc[:, :-1]
y = data.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

model = KNeighborsClassifier(n_neighbors=3)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Values:")
print(list(y_test))

print("\nPredicted Values:")
print(list(y_pred))

accuracy = accuracy_score(y_test, y_pred)
```

Output:

```
Actual Values:
['A', 'A', 'B', 'B', 'A', 'A']

Predicted Values:
['A', 'A', 'B', 'B', 'A', 'A']

Accuracy = 100.0 %
```

Result: The K-Nearest Neighbours (K-NN) algorithm was successfully implemented using Python. The classifier was trained with the given dataset, predicted the class labels for the test data, and achieved good classification accuracy.